Week2

Li-Hsin Chien

2026-03-05

1. Data

```
#library(faraway)
#data(gala, package="faraway")

gala<-read.table(file="gala.txt",header=T)

head(gala[,-2])
```

```
##   Species  Area Elevation Nearest Scruz Adjacent
## 1      58 25.09       346     0.6   0.6     1.84
## 2      31  1.24       109     0.6  26.3   572.33
## 3       3  0.21       114     2.8  58.7     0.78
## 4      25  0.10        46     1.9  47.4     0.18
## 5       2  0.05        77     1.9   1.9   903.82
## 6      18  0.34       119     8.0   8.0     1.84
```

2. Statistics

```
lmod <- lm(Species ~ Area + Elevation + Nearest + Scruz  + Adjacent, data=gala)
summary(lmod)
```

```
##
## Call:
## lm(formula = Species ~ Area + Elevation + Nearest + Scruz + Adjacent,
##     data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -111.679  -34.898   -7.862   33.460  182.584
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.068221  19.154198   0.369 0.715351
## Area        -0.023938   0.022422  -1.068 0.296318
## Elevation    0.319465   0.053663   5.953 3.82e-06 ***
## Nearest      0.009144   1.054136   0.009 0.993151
## Scruz       -0.240524   0.215402  -1.117 0.275208
## Adjacent    -0.074805   0.017700  -4.226 0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60.98 on 24 degrees of freedom
## Multiple R-squared:  0.7658, Adjusted R-squared:  0.7171
## F-statistic:  15.7 on 5 and 24 DF,  p-value: 6.838e-07
```

```
names(lmod)
```

```
##  [1] "coefficients"  "residuals"     "effects"       "rank"
##  [5] "fitted.values" "assign"        "qr"            "df.residual"
##  [9] "xlevels"       "call"          "terms"         "model"
```

```
names(summary(lmod))
```

```
##  [1] "call"          "terms"         "residuals"     "coefficients"
##  [5] "aliased"       "sigma"         "df"            "r.squared"
##  [9] "adj.r.squared" "fstatistic"    "cov.unscaled"
```

```
lmod$coefficients
```

```
##  (Intercept)         Area    Elevation      Nearest        Scruz     Adjacent
##  7.068220709 -0.023938338  0.319464761  0.009143961 -0.240524230 -0.074804832
```

```
summary(lmod)$coefficients
```

```
##                 Estimate  Std. Error      t value     Pr(>|t|)
## (Intercept)  7.068220709 19.15419782  0.369016796 7.153508e-01
## Area        -0.023938338  0.02242235 -1.067610554 2.963180e-01
## Elevation    0.319464761  0.05366280  5.953187968 3.823409e-06
## Nearest      0.009143961  1.05413595  0.008674366 9.931506e-01
## Scruz       -0.240524230  0.21540225 -1.116628222 2.752082e-01
## Adjacent    -0.074804832  0.01770019 -4.226216850 2.970655e-04
```

- sigma: $\hat{\sigma}$
- df: $(p, n-p, p*)$. p is the number of non-aliased coefficients, and $p*$ is the total number of coefficients.
- cov.unscaled: $(X^T X)^{-1}$

```
summary(lmod)$sigma
```

```
## [1] 60.97519
```

```
summary(lmod)$df
```

```
## [1]  6 24  6
```

```
deviance(lmod) # RSS
```

```
## [1] 89231.37
```

```
df.residual(lmod) # n-p
```

```
## [1] 24
```

```
sqrt(deviance(lmod)/df.residual(lmod)) #sigma_hat
```

```
## [1] 60.97519
```

```
summary(lmod)$cov.unscaled #(x'x)^-1
```

```
##               (Intercept)          Area     Elevation       Nearest
## (Intercept)  9.867829e-02  3.778242e-05 -1.561976e-04 -2.339027e-04
## Area         3.778242e-05  1.352247e-07 -2.593617e-07  1.294003e-06
## Elevation   -1.561976e-04 -2.593617e-07  7.745339e-07 -3.549366e-06
## Nearest     -2.339027e-04  1.294003e-06 -3.549366e-06  2.988732e-04
## Scruz       -3.760293e-04 -4.913149e-08  3.080831e-07 -3.821077e-05
## Adjacent     2.309832e-05  4.620303e-08 -1.640241e-07  1.424729e-06
##                     Scruz      Adjacent
## (Intercept) -3.760293e-04  2.309832e-05
## Area        -4.913149e-08  4.620303e-08
## Elevation    3.080831e-07 -1.640241e-07
## Nearest     -3.821077e-05  1.424729e-06
## Scruz        1.247941e-05 -1.958356e-07
## Adjacent    -1.958356e-07  8.426543e-08
```

```
x<-model.matrix(lmod)
solve(t(x)%*%x)
```

```
##               (Intercept)          Area     Elevation       Nearest
## (Intercept)  9.867829e-02  3.778242e-05 -1.561976e-04 -2.339027e-04
## Area         3.778242e-05  1.352247e-07 -2.593617e-07  1.294003e-06
## Elevation   -1.561976e-04 -2.593617e-07  7.745339e-07 -3.549366e-06
## Nearest     -2.339027e-04  1.294003e-06 -3.549366e-06  2.988732e-04
## Scruz       -3.760293e-04 -4.913149e-08  3.080831e-07 -3.821077e-05
## Adjacent     2.309832e-05  4.620303e-08 -1.640241e-07  1.424729e-06
##                     Scruz      Adjacent
## (Intercept) -3.760293e-04  2.309832e-05
## Area        -4.913149e-08  4.620303e-08
## Elevation    3.080831e-07 -1.640241e-07
## Nearest     -3.821077e-05  1.424729e-06
## Scruz        1.247941e-05 -1.958356e-07
## Adjacent    -1.958356e-07  8.426543e-08
```

3. Inferences

$$y = \beta_0 + \beta_1(Area) + \beta_2(Elevation) + \beta_3(Nearest) + \beta_4(Scruz) + \beta_5(Adjacent) + \varepsilon$$

F-test

$$H_0 : \beta_1 = \beta_2 = \beta_3 = \beta_4 = \beta_5 = 0$$

```
lmod <- lm(Species ~ Area + Elevation + Nearest + Scruz  + Adjacent, data=gala)
nullmod <- lm(Species~1, gala)

anova(nullmod, lmod)
```

```
## Analysis of Variance Table
##
## Model 1: Species ~ 1
## Model 2: Species ~ Area + Elevation + Nearest + Scruz + Adjacent
##   Res.Df    RSS Df Sum of Sq      F    Pr(>F)
## 1     29 381081
## 2     24  89231  5    291850 15.699 6.838e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
summary(lmod)$fstatistic
```

```
##    value    numdf    dendf
## 15.69941  5.00000 24.00000
```

```
(rss.null<-deviance(nullmod))
```

```
## [1] 381081.4
```

```
(rss<-deviance(lmod))
```

```
## [1] 89231.37
```

```
(df.null<-df.residual(nullmod))
```

```
## [1] 29
```

```
(df<-df.residual(lmod))
```

```
## [1] 24
```

```
(fstat<- (rss.null-rss)/(df.null-df)/(rss/df))
```

```
## [1] 15.69941
```

```
1-pf(fstat,df.null-df,df)
```

```
## [1] 6.837893e-07
```

Since the p-value of 6.8×10^{-7} is so small (<<0.05), this null hypothesis is rejected. As a group, the predictors have a statistically significant effect on the response variable.

Testing one predictor

We can test whether Area can be dropped from the full model by testing the hypothesis:

$$H_0 : \beta_1 = 0$$

```
lmods <- lm(Species ~ Elevation + Nearest + Scruz  + Adjacent, data=gala)
anova(lmods, lmod)
```

```
## Analysis of Variance Table
##
## Model 1: Species ~ Elevation + Nearest + Scruz + Adjacent
## Model 2: Species ~ Area + Elevation + Nearest + Scruz + Adjacent
##   Res.Df   RSS Df Sum of Sq      F Pr(>F)
## 1     25 93469
## 2     24 89231  1    4237.7 1.1398 0.2963
```

```
summary(lmod)$coefficients[2,3]^2
```

```
## [1] 1.139792
```

The p-value of 0.3 indicates that the null hypothesis cannot be rejected.

Testing a pair of predictors

$$H_0 : \beta_1 = \beta_5 = 0$$

```
lmods2 <- lm(Species ~ Elevation + Nearest + Scruz , data=gala)
anova(lmods2, lmod)
```

```
## Analysis of Variance Table
##
## Model 1: Species ~ Elevation + Nearest + Scruz
## Model 2: Species ~ Area + Elevation + Nearest + Scruz + Adjacent
##   Res.Df    RSS Df Sum of Sq      F  Pr(>F)
## 1     26 158292
## 2     24  89231  2     69060 9.2874 0.00103 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Confidence intervals for β

$$\hat{\beta}_i \pm t_{n-p}^{(\alpha/2)} se(\hat{\beta}_i)$$

Find the 95% confidence interval for β_1.

```
summary(lmod)
```

```
##
## Call:
## lm(formula = Species ~ Area + Elevation + Nearest + Scruz + Adjacent,
##     data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -111.679  -34.898   -7.862   33.460  182.584
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  7.068221  19.154198   0.369 0.715351
## Area        -0.023938   0.022422  -1.068 0.296318
## Elevation    0.319465   0.053663   5.953 3.82e-06 ***
## Nearest      0.009144   1.054136   0.009 0.993151
## Scruz       -0.240524   0.215402  -1.117 0.275208
## Adjacent    -0.074805   0.017700  -4.226 0.000297 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 60.98 on 24 degrees of freedom
## Multiple R-squared:  0.7658, Adjusted R-squared:  0.7171
## F-statistic:  15.7 on 5 and 24 DF,  p-value: 6.838e-07
```

```
qt(0.975,24)
```

```
## [1] 2.063899
```

```
-0.023938 + c(-1,1)*2.063899*0.022422
```

```
## [1] -0.07021474  0.02233874
```

```
confint(lmod)
```

```
##                   2.5 %      97.5 %
## (Intercept) -32.4641006 46.60054205
## Area         -0.0702158  0.02233912
## Elevation     0.2087102  0.43021935
## Nearest      -2.1664857  2.18477363
## Scruz        -0.6850926  0.20404416
## Adjacent     -0.1113362 -0.03827344
```